

QUERIES

1. A simple query of a single table.

Q: LIST THE CINEMA AND RATING OF ALL THE CINEMAS THAT HAVE A RATING ABOVE 3.0.

cinemaLocation	starRating
Blacktown	4.3
Broadway	4.3
Chatswood Mandarin	3.8
Chatswood Westfield	4.4
Eastgardens	4.2
(5 rows)	

A: `SELECT * FROM Cinema WHERE starRating > 3.0;`

2. A query which using the words “natural join”.

Q: LIST THE CINEMALOCATION, MOVIE NAME, RATING OF ALL CINEMAS THAT SHOW MOVIES THAT IS RATED M. ORDER BY MOVIE NAME.

cinemaLocation	movieName	movieYear	rating
Chatswood Mandarin	Doctor Strange in the Multiverse of Madness	2022	M
Chatswood Westfield	Doctor Strange in the Multiverse of Madness	2022	M
Broadway	Fall	2022	M
Eastgardens	The Batman	2022	M
Chatswood Mandarin	Thor: Love and Thunder	2022	M
Chatswood Westfield	Thor: Love and Thunder	2022	M
Broadway	Top Gun: Maverick	2022	M
(7 rows)			

A: `SELECT cinemaLocation, movieName, movieYear, rating FROM Cinema NATURAL JOIN CinemaToMovie NATURAL JOIN Movie WHERE rating = 'M' ORDER BY movieName;`

3. The cross-product equivalent to the “natural join” query above.

Q: Re-express the above natural join query as a cross-product.

A: `SELECT Cinema.cinemaLocation, Movie.movieName, Movie.movieYear, Movie.rating FROM Cinema, CinemaToMovie, Movie WHERE Cinema.cinemaLocation = CinemaToMovie.cinemaLocation AND Movie.movieName = CinemaToMovie.movieName AND Movie.movieYear = CinemaToMovie.movieYear AND rating = 'M' ORDER BY movieName;`

4. A query involving a “Group by”, perhaps also with a having.

Q: Using group by, create a query that produces the same result below.

HOYTS Branch	Rating	Number of Movies
Blacktown	4.3	2
Broadway	4.3	2
Chatswood Westfield	4.4	2
Eastgardens	4.2	2

(4 rows)

A: SELECT cinemaLocation AS "HOYTS Branch", starRating AS "Rating", COUNT(movieName) AS "Number of Movies" FROM Cinema NATURAL JOIN CinemaToMovie NATURAL JOIN Movie WHERE starRating > 4.0 GROUP BY CinemaLocation, starRating HAVING COUNT(*) > 1 ORDER BY cinemaLocation, starRating;

5. A query which uses a sub query.

Q: Use a query with a sub query to create the below result.

itemid	itemname	price
2	Regular Popcorn	10.00
3	Large Popcorn	11.00
7	Hawaiian Pizza	14.90
8	BBQ Meatlovers Pizza	14.90
12	Large Drink	9.70

(5 rows)

A: SELECT itemID, itemName, price FROM Item WHERE price > (SELECT AVG(price) FROM Item);

6. A cross product which cannot be implemented using the words "natural join" (e.g. self-join).

Q: SELF JOIN THAT REQUIRES CROSS PRODUCT NOTATION. CREATE THE BELOW RESULT.

HOYTS Member	Non-Member
784	523
784	764
476	523
476	764
892	523
892	764

(6 rows)

A: SELECT a.customerid AS "HOYTS Member", b.customerid AS "Non-Member" FROM Customer a, Customer b WHERE a.isMember = 'Y' AND b.isMember = 'N';